GaonMitra AI – Smart Farming Assistant

Empowering Indian Farmers through AI and Hindi Voice Support

By Neelam Nagar | B.Tech CSE (Prefinal Year), SKIT College, Jaipur | Duration: 8–9 Weeks

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The Challenges Farmers Face 🌾

Farmers struggle with multiple interconnected problems that directly impact their livelihoods and agricultural success:

- Farmers struggle to detect crop diseases on time.
- Weather information is unreliable or unavailable.
- Apps are only in English, not Hindi.
- They don't get **fair mandi prices**.
- Depend on **middlemen** for selling crops.
- Lack of awareness about **government subsidies**.

Impact: These problems lead to crop loss and reduced income.

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Our Solution – GaonMitra AI 🤖

A one-stop **AI-powered Hindi-speaking assistant** for farmers that transforms agricultural practices:

Detects Crop Diseases

Using advanced AI models

Real-Time Weather & Fertilizer Advice

Live data integration

Mandi Prices & Subsidy Info

Fair market information

Predicts Crop Price Trends

ML-based forecasting

Direct Buyer Connection

No middlemen involved

Bringing Artificial Intelligence to every Indian village!

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How GaonMitra AI Works ⚙️

The system follows a seamless flow from farmer input to actionable advice:

1

Farmer Input

Speaks in Hindi or uploads photo

2

AI Detection

Detects problem (disease, weather, or price)

3

Data Fetching

Retrieves from API, model, or database

4

Personalized Output

Hindi audio + text advice

Core Modules:

- Voice Recognition
- Crop Disease AI Model
- APIs (Weather, Mandi)
- ML Price Predictor
- Buyer & Subsidy Database

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Technologies Used 💻

Category	Tools / Libraries
Language	Python
AI / ML	TensorFlow, scikit-learn
Image Processing	OpenCV
Voice & NLP	SpeechRecognition, gTTS, googletrans
APIs	OpenWeatherMap, Agmarknet (Data.gov.in)
Data Storage	SQLite / CSV
Visualization	Matplotlib
IDE	Jupyter Notebook / VS Code

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Features of GaonMitra AI 🌱

1

Crop Disease Detection

CNN Model for accurate identification

2

Weather & Fertilizer Advice

Live API integration

3

Hindi Voice Assistant

Speech + gTTS support

4

Mandi Bhav Module

Market Price Information

5

Government Subsidy Info

PM-Kisan, PMFBY, and more

6

Smart Price Advisor

ML-based Price Prediction

7

Direct Farmer–Buyer Connect

No Middlemen involved

All features available through Hindi voice & text.

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Smart Price & Market Intelligence 💰

The Smart Price Advisor Module empowers farmers with data-driven selling decisions:

7-Day Price Prediction

ML regression models forecast crop prices for the next week

Multi-City Comparison

Compares mandi prices from nearby cities

Optimal Selling Strategy

Suggests best time and place to sell crops

Profit Maximization

Helps farmer earn 20–30% more profit

Machine Learning meets Agri-Economics!

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Ending the Middleman System 🔗

The Direct Buyer Connect feature revolutionizes how farmers sell their crops:

How It Works

- Farmers can view list of local traders / industries
- See crop type, price, and contact details
- Connect directly via call or message
- Encourages fair trade & transparency

Empowering farmers to sell directly & earn more!

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Development Roadmap & Impact 🧭

The comprehensive 9-week development plan ensures systematic progress:

Week	Task	Outcome	Milestone
1	Setup & data collection	Dataset + environment ready	Foundation
2–3	Train Crop Disease CNN	Model accuracy ~90%	Core AI
4	Add Weather & Fertilizer API	Real-time updates	Live Data
5	Build Hindi Voice Assistant	Voice + text support	Language
6	Add Mandi Bhav & Subsidy Module	Price & scheme data	Market Info
7	Train Smart Price Predictor	Price trend forecasting	Analytics
8	Add Buyer Connect System	Direct trade option	Connectivity
9	Integration + Testing	Full AI system ready for demo	Complete

Project Results & Future Vision ✨

Current Achievements

- Detects crop diseases accurately
- Predicts crop price trends
- Speaks to farmers in Hindi
- Shows mandi prices & subsidies
- Suggests direct buyers nearby

Future Enhancements

- Convert to Android App (Kotlin / Flutter)
- Use IoT sensors for real-time soil data
- Add Drone image detection for large farms
- Use WhatsApp chatbot for farmers without apps
- Store farmer history & analytics in Firebase

Impact on Rural India 🇮🇳

Farmers can now:

- Detect diseases early
- Get advice in their own language
- Know real mandi prices
- Contact buyers directly
- Increase profit and confidence

> **"Technology se Samruddh Kisan" — AI for Rural Empowerment!**

Thank You 🙏

Let's make India's villages smarter together.

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